

Anuran swingers: misdirected mating attempts occurred early during anuran diversification

François Brischoux, Léa Lorrain-Soligon

▶ **To cite this version:**

François Brischoux, Léa Lorrain-Soligon. Anuran swingers: misdirected mating attempts occurred early during anuran diversification. Biological Journal of the Linnean Society, 2023, blad108, ⟨10.1093/biolinnean/blad108⟩. ⟨hal-04196592⟩

HAL Id: hal-04196592

https://hal.science/hal-04196592v1

Submitted on 15 Nov 2024

HAL is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



HAL Authorization

Anuran swingers: misdirected mating attempts have occurred early during the anuran diversification

François Brischoux[1], Léa Lorrain-Soligon[1]

1. Centre d'Etudes Biologiques de Chizé, UMR7372 CNRS-La Rochelle Université, 79360 Villiers en Bois, France

Running head: Misdirected matings in anurans

Correspondence:

F. Brischoux (francois.brischoux@cebc.cnrs.fr)

CEBC UMR7372 CNRS-ULR, 79360 Villiers en Bois, France

Abstract.

Promiscuity, explosive breeding and male-biased operational sex-ratios can affect the strength of male selectivity and may play major roles in the expression of misdirected matings (with conspecific males, other species, corpses or objects) in anuran amphibians. Yet, misdirected amplexus occur in multiple species diverging from these reproductive traits, suggesting that the expression of such behaviour is widespread, and may have occurred early during the anuran diversification. Using two methods of ancestral-trait-reconstruction which predict unknown states, we found a very high probability that most anuran genera express misdirected amplexus, suggesting that this by-product of anuran reproductive strategies is likely to have occurred early during the anuran diversification. The persistence of misdirected amplexus suggests that its infrequent occurrence may outweigh individual costs of breeding attempts with unfit mates. We found a recent exponential increase of reported observations of misdirected amplexus, which, along with increased research effort and publication rate, may reveal an effect of environmental perturbations known to promote the occurrence of these behaviours.

Key-words: amphibians, ancestral state reconstruction, reproduction, Hidden State Prediction, Generalized Hidden Markov Model, heterospecific matings

Introduction

In sexually reproducing organisms, a universal aspect of reproduction involves the ability to recognize conspecifics of the opposite sex to produce offspring and thus improve chances of reproductive success and overall fitness (Emlen & Oring 1977; Mayr 2017). Although hybridization is possible between some species and can promote evolutionary diversification (Dowling & Secor, 1997), incompatibility between heterospecifics usually leads to reproductive failure (Hettyey & Pearman 2003; Kudo & Kasagi 2005). This generates strong selection to evolve traits that promote mechanisms of reproductive isolation for matings to occur exclusively between conspecifics (Winter & Rollenhagen 1990; Rodríguez et al. 2004). Yet, heterospecific matings are often observed across diverse phylogenetic groups, and have various consequences, including species displacement, spatial or temporal segregation, changes in life history parameters, and reproductive character displacement (Gröning & Hochkirch 2008). Studying these situations have offered opportunities for investigations into mate selection (Pfennig 2007), mating signals (Wyman et al. 2011), genetic incompatibility (Kyogoku & Wheatcroft 2020), reproductive isolation (Vega-Sánchez et al. 2022), reproductive interference (Kyogoku 2015), hybridization (Hartke & Rosengaus 2011), and the role of geographic distribution (i.e., allopatry *versus* sympatry) in these evolutionary processes (Verrell 1994; Ruokolainen & Hanski 2016; Kyogoku & Kokko 2020).

Heterospecific (misdirected) mating attempts have been commonly observed among anuran amphibians (Serrano et al. 2022). This is presumably due to the specific

reproductive biology of this taxonomic group. Reproduction in anurans involves amplexus, the mating position in which a male grasps a female prior to and until the completion of oviposition and fertilization (Wells 2007). Amplexus can last for a significant duration (up to several days, Wells 1977; Wells 2007) and are relatively conspicuous as compared to secretive or short matings that are difficult to observe in many other taxa. The conspicuousness of amplectant pairs in anurans has led to numerous observations of misdirected amplexus in this highly diverse amphibian lineage (Serrano et al. 2022). These misdirected amplexus can involve conspecific males (Lago Londero et al. 2018, Mollov et al. 2010), anuran heterospecifics (both male or female individuals), but also include numerous observations of amplexus with phylogenetically disparate species (e.g., caudates, fish, turtles, Simović et al. 2014, Zimić et al. 2018), dead individuals (i.e., necrophilia, Pintanel et al. 2021), and inanimate objects (e.g., Mollov et al. 2010, Sodré et al. 2018).

Both the geographic and phylogenetic uneven distributions of misdirected amplexus among anurans (Serrano et al. 2022) - although presumably linked to taxonomic bias in research intensity - have led to several hypotheses in order to explain the occurrence of such atypical behaviour (Serrano et al. 2022). These hypotheses involve the role of breeding strategies (explosive *versus* prolonged breeding, Fouquet et al. 2021), the role of scramble competition (Hettyey et al. 2014; Hettyey & Pearman 2003; Schmeller et al. 2005), the lack of or alteration of male-female or species discrimination (Schmeller et al. 2005; Cheong et al. 2008; Simmons & Narins 2018), the influence of other traits such as longevity or morphology (Schmeller et al. 2005),

or, potentially, a combination thereof (Serrano et al. 2022). In such conditions, it would be advantageous for males to arbitrarily be attracted to – and thus clasp - any female-looking object in order to increase mating probabilities (Serrano et al. 2022). This is supported by the geographic distribution of species in which misdirected amplexus were more frequently recorded (Neartic and Palaeartic, Serrano et al. 2022). In these areas, suitable reproductive conditions are restricted in time, favouring explosive breeding thereby promoting higher occurrence of indiscriminate amplexing behavior (Pearl et al., 2005; Canestrelli et al., 2017). In turn, this may help to explain the phylogenetic distribution of misdirected amplexus, which occur more frequently in three anuran families (Bufonids, Ranids and Hylids, Serrano et al. 2022) in which explosive breeding and high level of male-male competition are widespread (Gaitonde et al. 2016, Bell et al. 2017; Pearl et al. 2007).

Communal, explosive breeding strategies are likely to be a primary cause of misdirected amplexus. However, misdirected amplexus have been observed worldwide in species with contrasting breeding strategies, including those with prolonged breeding (e.g. Höbel 2005; Ferreira et al. 2019) and asynchronysed mating periods (Mollov et al. 2010), in which mating occurs individually on land or on trees rather than communally in water (Ferreira et al. 2019; Lu et al. 2016; Vásquez-Cruz et al. 2019), with amplexus initiated by female signals (e.g., Wells 1977), and in which mate selection occurs through acoustic cues (Gerhardt 1994; Nityananda & Bee 2011). These exceptions led to the intriguing possibility that the expression of misdirected amplexus is ubiquitous across anurans. In turn, this suggests that the expression of

atypical misdirected amplexus - and thus conditions promoting this behaviour – may have occurred early during the anuran diversification. Importantly, this idea does not invalidate previously highlighted hypotheses but rather root these hypotheses more deeply at the basis of anuran phylogeny, prior to diversification processes that induced variations from the putative ancestral reproductive characteristics of this taxonomic group (Gomez-Mestre et al. 2012; Carvajal-Castro et al., 2020).

In this study, we tested whether misdirected amplexus occurred early during the anuran diversification on a comprehensive dataset of published observations in anurans using state-of–the-art ancestral state reconstruction approaches which allow to predict unknown states, because these data are relatively scarce across the whole anuran diversity. We predicted that, as observations of misdirected amplexus are distributed across all continents (see Serrano et al. 2022) and are observed in species with contrasted life history traits this by-product of anuran reproductive strategies is likely to have occurred early during the anuran diversification.

Material and methods

Data collection

We used the dataset assembled by Serrano et al. (2022) along with missing references and additional data published since then (Appendix A and B). We followed the same methodology as described by Serrano et al. (2022). Briefly, we searched articles mentioning misdirected amplexus using key words such as 'frog', 'toad', 'amphibian', 'anuran', 'anurans', 'Anura' and 'Amphibia' combined with 'amplexus' 'abnormal', 'unusual', 'interspecific' and 'heterospecific'. We did not check for photographic records shared via social media, but did keep those assembled by Serrano et al. (2022). We deliberately excluded from our analyses (see below) records of necrophilia (Davian behaviour) and amplexus with inanimate objects as these events may represent extreme and atypical situations. From this list of species, we updated taxonomic identity using Jetz & Pyron (2018) and Frost (2021).
In total, our database included 280 records of misdirected (interspecific) amplexus, representing 159 species across 70 genera (Appendices A and B).

Statistical analyses

The structure of our dataset led to several analytical constraints for ancestral state reconstruction. Indeed, the absence of published records of misdirected amplexus in numerous species does not necessarily mean that such atypical behaviour does not occur in these species. As a consequence, we could not univocally assess the absence of this behaviour in species in which misdirected amplexus has not been observed or published to date. This leads to the existence of unknown states in a large number of

species. To take into account such complexity, we used up-to-date analytical approaches to compute ancestral states: the Hidden State Prediction (HSP) method using the package castor (v. 1.7.3) (Louca & Doebeli, 2017), and the Generalized Hidden Markov Model (HMM) method using the package corHMM (v. 2.8) (Beaulieu et al. 2022), with an ER model (Equivalent Rates, AICc=15.277, compared to AICc=16.475 for the All Rates Different [ARD] model), and no fixed nodes probability. Based on a collection of known characters in a limited number of species, these methods allow for unknown character states to be predicted at the tips of a phylogenetic tree as well as for intermediate nodes. For the same reasons (impossibility to obtain univocal data demonstrating that a given species does not express misdirected amplexus), we set a fixed probability of 0 for "no misdirected amplexus possible" in the few species (genera *Adelphobates*, *Andinobates*, *Dendrobates*, and *Oophaga*) that are known to have lost amplexus for reproduction (Carvajal-Castro et al., 2020). Our rational was that if these species do not display amplexus during reproduction, they are arguably unlikely to display misdirected amplexus. Using this approach, a single conflict emerged for the genus *Nanorana*, which has been classified as "no amplexus" in Carvajal-Castro et al. (2020) but for which we found a published observation of an amplexus (Lu et al., 2016). In this case, we kept species of this genus in the category "performing amplexus".

We used TimeTree of Life (Kumar et al. 2022) to produce the phylogenetic tree used for our analyses. The anuran phylogenetic tree available in TimeTree of Life (accessed on 15/04/2022) contained 3505 species (Kumar et al. 2022), from which 159

species have been observed completing misdirected amplexus. Thus, 20 species were classified as "no misdirected amplexus possible" and 3324 species with unknown states. Consequently, we set the genus - rather than the species – as the taxonomic unit to perform our analyses. This approach allowed us to reduce the frequency of unknown states to 82.3% (N= 67 genera) rather than 94.8% using species as the taxonomic unit.

Using the tip states estimated from both methods, we illustrated the accumulation through time of lineages (lineages-through-time plots) in each of the mapped states, allowing us to determinate the time of diversification (Helmestter et al., 2022), using the package *phytools* (v. 1.1.7) (Revell 2012). We computed ER (Equivalent Rates) models, through selection via AICc (AICc=20.783 compared to AICc=22.767 for the All Rates Different [ARD] models), with 100 iterations.

All data analyses were performed using R 3.6.3 (R Core Team, 2020) and Rstudio v1.1.419.

Results

Ancestral trait reconstruction

Both statistical approaches (HSP and HMM) yielded similar results, but with small differences.

Using HMM, the probability that misdirected amplexus occurred at the ancestral node of the current anuran phylogeny was very high (0.998, Figure 1). Overall, across all nodes in the phylogenetic tree, the mean probability that misidirected amplexus occurred was also very high (0.962±0.163). The few nodes for which this probability was relatively low were highly clustered (Figure 1) and corresponded to instances when species lacked amplexus for breeding (an information we used to set "no misdirected amplexus possible", see methods). These nodes represented 4.25% (N=17) of all nodes with probabilities that misidirected amplexus occurred ranging from $5.556E^{-7}$ to 0.419 (0.204±0.168). All other nodes had probabilities higher than 0.973.

Using HSP, the probability that misdirected amplexus occurred at the ancestral node of the current anuran phylogeny was very high (0.976, Appendix C). Overall, across all nodes in the phylogenetic tree, the mean probability that misdirected amplexus occurred was also very high (0.944±0.145). In contrast with the HMM method, the distribution of probabilities was multimodal. Very few nodes had a probability equal to 0 (N=6, 1.5% of all nodes) and, similarly to the HMM method, these nodes were highly clustered (Appendix C) and corresponded to instances when species lacked amplexus for breeding. There was a second group of nodes with intermediate probabilities ranging from 0.648-0.878 (0.770±0.059, N=63, 15.7% of all nodes,

Appendix C). All other nodes had probabilities higher than 0.946, among which 290 had probabilities equal to 1 (Appendix C).

Lineage through time

This additional set of analyses was convergent with ancestral traits reconstruction. Using tip states estimated from the HMM approach, this analysis dated the apparition of misdirected amplexus at the oldest node of the phylogenetic tree we used (220 Mya, Figure 2), while the loss of this character (i.e., the emergence of species that do not perform misdirected amplexus) was much more recent (60 Mya, Figure 2). The results from the tip states estimated from the HSP method yielded similar results, except for the date of misdirected amplexus loss which was estimated to emerge more recently (30 Mya, Appendix D).

Discussion

Ancestral state reconstructions indicated that there is a high probability that most anuran species exhibit misdirected amplexus. More importantly, these analyses also indicated that these atypical mating attempts may have occurred early during the anuran diversification, suggesting that the conditions promoting this behaviour (i.e., explosive breeding and scramble male-male competition because of highly biased operational sex-ratios) may also have occurred in anurans prior to the remarkable diversification of reproductive modes in these taxa (Gomez-Mestre et al., 2012; Crump, 2015; Carvajal-Castro et al., 2020). In support to this finding, our analyses dated the emergence of this behaviour at the earliest node in our dataset (~220 Mya). Although it is plausible that the emergence of this behaviour can be older or younger given the uncertainties in our dataset, it is noteworthy that this period is situated after the split between anurans and caudates, but before the separation of the Gondwana (San Mauro, 2010), which dovetails relatively well with the fact that misdirected amplexus have been observed worldwide (see Serrano et al. 2022). Taken together, these elements reinforce the notion that these atypical behaviours may have occurred early during the anuran diversification, rather than linked to multiple independent origins of a reproductive dead-end. Similarly, our results (HMM) suggest that the loss of misdirected amplexus occurred ~ 60 Mya a period during which a simultaneous diversification of anuran clades occurred (Feng et al. 2017).

Previous studies have investigated ancestral reproduction modes of anurans. Gomez-Mestre et al. (2012) have demonstrated that aquatic breeding, egg-laying and

larval development are all ancestral traits in anurans despite the high diversity of reproductive modes exhibited currently (e.g., terrestrial reproduction, direct development). More recently, Carvajal-Castro et al. (2020) have shown that amplexus has been conserved across the evolutionary history of anuran despite several evolutionary transitions in amplexus type. In combination with current hypotheses about the conditions that cause atypical misdirected amplexus (Schmeller et al. 2005; Serrano et al. 2022), our results provide further insights to ancestral anurans reproductive modes. Indeed, if misdirected amplexus is promoted by explosive breeding with scramble male-male competition (Serrano et al. 2022), then our analyses suggest that such conditions were presumably basal to the anuran diversification. Unfortunately, it was not possible to test for covariation patterns between the occurrence of misdirected amplexus and other reproductive traits that may maintain such behaviour (i.e., explosive breeding, rival density, biased operational sex-ratio, aquatic breeding) because we could not robustly assess the univocal presence or absence of misdirected amplexus in groups for which this behaviour has not been observed to date. We believe that increasing natural history observations will be crucial to circumvent such caveat and to formally assess the reproductive traits that promote heterospecific matings in anurans. Our dataset showed that the frequency of observations of misdirected amplexus in anurans has exponentially increased since the early 2000's (Appendix E) which is promising to test for covariation patterns between the occurrences of misdirected amplexus and other reproductive traits. Photographic posts on social media or naturalist applications (such as iNaturalist) can play a key role in observations of unusual

behaviors and help scientists to build more comprehensive datasets (Edwards et al. 2021; Qi et al. 2019).

Although hybridization can promote evolutionary diversification (Dowling & Secor, 1997) – a process that has been found in anurans (Kraus 2015, Pabijan et al. 2020) - incompatibility between heterospecifics usually led to reproductive failure (Hettyey & Pearman 2003; Kudo & Kasagi 2005). Why then misdirected amplexus can be so widespread across anurans? We believe that the answer may lay in the relative individual *versus* populational (or indeed species) costs of the reproductive failure induced by misdirected amplexus. Indeed, under the conditions that may promote misdirected matings (explosive breeding with scramble male-male competition), it would be advantageous for males to arbitrarily be attracted to – and then clasp - any female-looking object in order to increase mating probabilities (Serrano et al. 2022). In such circumstances, increased mating probabilities with conspecifics are likely to outweigh the individual costs of atypical indiscriminate breeding attempts with unfit mates (including other anuran species, phylogenetically disparate species, dead bodies or inanimate objects, Mollov et al. 2010; Pintanel et al. 2021; Simović et al. 2014; Sodré et al. 2018, Shin et al. 2020). For example, amplexus attempts with conspecific of the same sex is often observed across anurans (Lago Londero et al. 2018, Mollov et al. 2010, a behaviour that we did not include in our analysis), and specific strategies (e.g., 'release call', Schmidt 1972) have evolved in order to avoid reproductively useless homosexual amplexus (Toledo et al. 2015). Overall, fierce competition in small water bodies during short breeding seasons are likely to

promote relatively weak mate discrimination but, in turn, promiscuity induced by explosive breeding may decrease the costs of infrequent misdirected amplexus – and thus reproductive failure - at the population level. In addition, in areas where species promiscuity is high and misdirected amplexus frequent, females have developed strategies to increase the likelihood of being released by an heterospecific male (but with partial loss of fitness, Hettyey et al. 2009) and/or to adjust mate preference to reduce hybridization risk (Bell & Zamudio 2012, Calabrese & Pfennig 2022). Theoretical modelling approaches could be useful to test for this hypothesis. Such approach would allow to test the relative contribution of mate and rival density (i.e., operational sex-ratio) but also the number of heterospecific species breeding simultaneously, and the duration of breeding season (i.e., explosive *versus* prolonged breeding).

There are some caveats to our investigations, notably linked to the fact that we based our analyses on sparse information. That is, we could not univocally assess the absence of misdirected amplexus in species in which such behaviour has not been observed to date. Thus, we had to rely on indirect information to identify species in which misdirected amplexus were unlikely to occur (i.e., species in which amplexus have been lost during diversification, Carvajal-Castro et al., 2020). In addition, we performed our analyses with a relatively low number of species in which misdirected amplexus have been observed relative to the whole anuran diversity (N=159 over about 6700 species [Jetz & Pyron 2018], representing 2.4% of the anuran current diversity). As a consequence, we used two different statistical approaches developed

to infer unknown states at the tips and nodes of phylogenetic trees (HSP and HMM, see methods). Despite the usefulness of these analytical tools to decipher evolutionary pathways when available data are scarce, we emphasize that the low number of taxa in which misdirected amplexus have been actually observed may have hampered our ability to detect putative independent transitions across the anuran phylogeny as those found for amplexus types (Carvajal-Castro et al., 2020). Despite these caveats, we think that the phylogenetic distribution of species in which misdirected amplexus have been observed across anurans (i.e., these observations were not clustered within restricted families or genus, Figure 1, Appendices A, C) as well as the geographic distribution of these observations (that have been found worldwide, Serrano et al. 2022) strengthen our results.

Recent studies have highlighted that current environmental perturbations linked to anthropogenic activities and associated global change (e.g., droughts, habitat modifications, anthropogenic noise, invasive species) may promote the occurrence of misdirected amplexus (Cory & Manion, 1955; Pearl et al., 2005; D'Amore et al., 2009; Canestrelli et al., 2017; Candolin, 2019; Simmons & Narins 2018; see Serrano et al. 2022). Thus, it is plausible that the recent exponential increase in the frequency of observations of misdirected amplexus in anurans (Appendix E) is linked to such processes in combination to increased research effort and publication rate. If such hypothesis holds true, anthropogenic perturbations of mate discrimination in reproducing anuran may ultimately have stronger impacts on the persistence of these endangered taxa.

To conclude, our results suggest that misdirected amplexus, which could led to reproductive failure, are widespread across the anuran phylogeny and may have arisen prior to anuran diversification. Although we suggest that the persistence of such atypical behaviour during evolutionary time may have been possible because infrequent occurrence within populations may outweigh individual costs of indiscriminate breeding attempts with unfit mates, the current exponential increase of this behaviour may reveal a worrisome effect of environmental perturbations linked to global change. Increased investigations and observations of these behaviors (either in scientific literature or through citizen science) could allow to test for these different hypotheses.

Acknowledgments

We thank the CNRS for funding. LLS was supported by a PhD grant from La Rochelle Université. We warmly thank Jean-Pierre Vacher for discussions on the hypothesis tested in this MS, as well as for sharing his expertise in ancestral state reconstruction and in anuran Natural History. We thank Filipe C. Serrano, Juan C. Díaz-Ricaurte and Marcio Martins for making their dataset available which was very helpful to complement our initial collection and two anonymous Referees for providing insightful comments.

References

Beaulieu J, O'Meara B, Oliver J, Boyko J. 2022. _corHMM: Hidden Markov Models of Character Evolution_. R package version 2.8.

Bell RC, Webster GN, Whiting MJ. 2017. Breeding biology and the evolution of dynamic sexual dichromatism in frogs. *Journal of evolutionary biology* 30(12), 2104-2115.

Bell RC, Zamudio KR. 2012. Sexual dichromatism in frogs: natural selection, sexual selection and unexpected diversity. *Proc. R. Soc. B*. 279:4687–4693

Calabrese GM, Pfennig KS. 2022. Females alter their mate preferences depending on hybridization risk. *Biol. Lett*. 18:20220310

Candolin U. 2019. Mate choice in a changing world. *Biological Reviews* 94:1246–1260.

Canestrelli D, Bisconti R, Chiocchio A, Maiorano L, Zampiglia M, Nascetti G. 2017. Climate change promotes hybridisation between deeply divergent species. *PeerJ* 5:e3072.

Carvajal-Castro JD, López-Aguirre Y, Ospina-L. AM, Santos JC, Rojas B, Vargas-Salinas F. 2020. Much more than a clasp: evolutionary patterns of amplexus diversity in anurans. *Biological Journal of the Linnean Society* 129:652–663.

Cheong S, Sung H, Park S. 2008. Inability of mate and species recognition by male Asian toads, *Bufo gargarizans*. *Animal Cells and Systems* 12: 93–96.

Cory L, Manion JJ. 1955. Ecology and hybridization in the genus *Bufo* in the Michigan-Indiana region. *Evolution* 1955:42–51.

Crump ML. 2015. Anuran reproductive modes: evolving perspectives. *Journal of Herpetology* 49(1), 1-16.

D'Amore A, Kirby E, Hemingway V. 2009. Reproductive interference by an invasive species: an evolutionary trap. *Herpetological Conservation and Biology* 4:325–330.

Dowling TE, Secor CL. 1997. The role of hybridization and introgression in the diversification of animals. *Annual review of Ecology and Systematics* 593-619.

Edwards T, Jones CB, Perkins SE, Corcoran P. 2021. Passive citizen science: The role of social media in wildlife observations. *Plos one*, 16(8), e0255416.

Emlen ST, Oring LW. 1977. Ecology, sexual selection, and the evolution of mating systems. *Science* 197(4300), 215-223.

Feng Y-J, Blackburn DC, Liang D, Hillis DM, Wake DB, Cannatella DC, Zhang P. 2017. Phylogenomics reveals rapid, simultaneous diversification of three major clades of Gondwanan frogs at the cretaceous–Paleogene boundary. *Proceedings of the National Academy of Sciences of the United States of America* 114, E5864–E5870

Ferreira FCL, Kloss TG, Soares TS, Tonini JFR, Mônico AT, Ferreira RB. 2019. New records of interspecific amplexus in Neotropical anurans. *Herpetology Notes* 12, 705-708.

Fouquet A, Tilly T, Pašukonis A, Courtois EA, Gaucher P, Sebastian Ulloa J, Sueur J. 2021. Simulated chorus attracts conspecific and heterospecific Amazonian explosive breeding frogs. *Biotropica* 53(1), 63-73.

Frost DR. 2021. Amphibian species of the world: an online reference. Version 6.1. Available at: http://research.amnh.org/herpetology/amphibia/index.html

Gaitonde N, Giri V, Kunte K. 2016. 'On the rocks': reproductive biology of the endemic toad *Xanthophryne* (Anura: Bufonidae) from the Western Ghats, India. *Journal of Natural History* 50(39-40), 2557-2572.

Gerhardt HC. 1994. Reproductive character displacement of female mate choice in the grey treefrog, *Hyla chrysoscelis*. *Animal Behaviour* 47(4), 959-969.

Gomez-Mestre I, Pyron RA, Wiens JJ. 2012. Phylogenetic analyses reveal unexpected patterns in the evolution of reproductive modes in frogs. *Evolution* 66:3687-3700

Gröning J, Hochkirch A. 2008. Reproductive interference between animal species. *The Quarterly review of biology* 83(3), 257-282.

Hartke TR, Rosengaus RB. 2011. Heterospecific pairing and hybridization between *Nasutitermes corniger* and *N. ephratae*. *Naturwissenschaften* 98(9), 745-753.

Helmstetter AJ, Glemin S, Käfer J, Zenil-Ferguson R, Sauquet H, de Boer H, ... & Condamine FL. 2022. Pulled diversification rates, lineages-through-time plots, and modern macroevolutionary modeling. *Systematic Biology* 71(3), 758-773.

Hettyey A, Pearman PB. 2003. Social environment and reproductive interference affect reproductive success in the frog *Rana latastei*. *Behavioral Ecology* 14(2), 294-300.

Hettyey A, Baksay S, Vagi B, Hoi H. 2009. Counterstrategies by female frogs to sexual coercion by heterospecifics. *Animal Behaviour* 78(6), 1365-1372.

Hettyey A, Vági B, Kovács T, Ujszegi J, Katona P, Szederkényi M, ... Hoi H. 2014. Reproductive interference between *Rana dalmatina* and *Rana temporaria* affects reproductive success in natural populations. *Oecologia* 176(2), 457-464.

Höbel G. 2005. *Rana clamitans* (Green Frog) and *Rana catesbeiana* (American Bullfrog). *Herpetological Review* 36:439–440.

Jetz W, Pyron RA. 2018. The interplay of past diversification and evolutionary isolation with present imperilment across the amphibian tree of life. *Nature Ecology and Evolution* 2:850–858.

Kooijman SA, Lika K. 2014. Resource allocation to reproduction in animals. *Biological Reviews* 89(4), 849-859.

Kraus F. 2015. Impacts from invasive reptiles and amphibians. *Annual Review of Ecology, Evolution, and Systematics* 46, 75-97.

Kudo G, Kasagi T. 2005. Microscale variations in the mating system and heterospecific incompatibility mediated by pollination competition in alpine snowbed plants. *Plant Species Biology* 20(2), 93-103.

Kumar S, Suleski M, Craig JE, Kasprowicz AE, Sanderford M, Li M, Stecher G, Hedges SB. 2022. TimeTree 5: An Expanded Resource for Species Divergence Times. Molecular Biology and Evolution, DOI: 10.1093/molbev/msac174.

Kyogoku D. 2015. Reproductive interference: Ecological and evolutionary consequences of interspecific promiscuity. *Population Ecology* 57, 253–260.

Kyogoku D, Kokko H. 2020. Species coexist more easily if reinforcement is based on habitat preferences than on species recognition. *J. Anim. Ecol*. 89, 2605–2616

Kyogoku D, Wheatcroft D. 2020. Heterospecific mating interactions as an interface between ecology and evolution. *Journal of Evolutionary Biology* 33(10), 1330-1344.

Lago Londero JE, dos Santos Feltrin R, da Rocha MC, Schuch AP, dos Santos MB 2018. Interspecific amplexus between two treefrogs of the genus *Boana* Gray, 1825 (Anura: Hylidae) in captivity: male-female and male-male pairings. *Herpetology Notes* 11, 413-415.

Louca S, Doebeli M. 2017. “Efficient comparative phylogenetics on large trees.” _Bioinformatics_. doi:10.1093/bioinformatics/btx701

Lu X, Ma X, Fan L, Hu Y, Lang Z, Li Z, Fang B, Guo W. 2016. Reproductive ecology of a Tibetan frog *Nanorana parkeri* (Anura: Ranidae). *Journal of Natural History* 50:2769–2782.

Mayr E. 2017. Sexual selection and natural selection. In Sexual selection and the descent of man (pp. 87-104). Routledge.

Mollov IA, Popgeorgiev GS, Naumov BY, Tzankov ND, Stoyanov AY. 2010. Cases of abnormal amplexus in anurans (Amphibia: Anura) from Bulgaria and Greece. *Biharean Biologist* 4(2), 121-125.

Nityananda V, Bee MA. 2011. Finding your mate at a cocktail party: frequency separation promotes auditory stream segregation of concurrent voices in multi-species frog choruses. *PLoS One* 6(6), e21191.

Pabijan M, Palomar G, Antunes B, Antoł W, Zieliński P, Babik W. 2020. Evolutionary principles guiding amphibian conservation. *Evolutionary Applications* 13(5), 857-878.

Pearl CA, Hayes MP, Haycock R, Engler JD, Bowerman J. 2005. Observations of interspecific amplexus between western North American ranid frogs and the introduced American bullfrog (*Rana catesbeiana*) and an hypothesis concerning breeding interference. *The American Midland Naturalist* 154:126–134.

Pearl CA, Adams MJ, Wente WH. 2007. Characteristics of Columbia spotted frog (*Rana luteiventris*) oviposition sites in northeastern Oregon, USA. *Western North American Naturalist* 67(1), 86-91.

Pfennig KS. 2007. Facultative mate choice drives adaptive hybridization. *Science* 318(5852), 965-967.

Pianka ER. 1976. Natural selection of optimal reproductive tactics. *American Zoologist* 16(4), 775-784.

Pintanel P, Obando-Moreno G, Merino-Viteri A. 2021. Necrophiliac behaviour in the recently described species *Scinax tsachila* (Anura: Hylidae), with a review of necrophilia in amphibians. *Neotropical Biodiversity* 7(1), 53-56.

Qi L, Li J, Wang Y, Gao X. 2019. Urban observation: Integration of remote sensing and social media data. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 12(11), 4252-4264.

R Core Team. 2020. R: A language and environment for statistical computing. R Foundation for Statistical Computing. Austria: Vienna. https://www.R-project.org/.

Revell LJ. 2012. phytools: an R package for phylogenetic comparative biology (and other things). *Methods in ecology and evolution* (2), 217-223.

Rodríguez RL, Sullivan LE, Cocroft RB. 2004. Vibrational communication and reproductive isolation in the *Enchenopa binotata* species complex of treehoppers (Hemiptera: Membracidae). *Evolution* 58(3), 571-578.

Ruokolainen L, Hanski I. 2016. Stable coexistence of ecologically identical species: conspecific aggregation via reproductive interference. *J. Anim. Ecol*. 85, 638–647

San Mauro D. 2010. A multilocus timescale for the origin of extant amphibians. *Molecular Phylogenetics and Evolution* 56, 554-561.

Schmeller DS, O'Hara R, Kokko H. 2005. Male adaptive stupidity: male mating pattern in hybridogenetic frogs. *Evolutionary Ecology Research* 7(7), 1039-1050.

Schmidt RS. 1972. Release calling and inflating movements in anurans. *Copeia* 1972(2):240–245

Serrano FC, Díaz-Ricaurte JC, Martins M. 2022. Finding love in a hopeless place: A global database of misdirected amplexus in anurans. *Ecology* 2022:e3737.

Shin Y, Ambu J, Borzée A. 2020. Additional observations on heterospecific amplexus in Asiatic toads (Anura: Bufonidae: *Bufo gargarizans*) in the Republic of Korea. *Herpetology Notes* 13, 411-413.

Simmons AM, Narins PM. 2018. Effects of anthropogenic noise on amphibians and reptiles. In Effects of anthropogenic noise on animals (pp. 179-208). Springer, New York, NY.

Simović A, Anderson N, Andelković M, Gvozdenović S, Dorđević S. 2014. Unusual amplexuses between anurans and caudates. *Herpetology Notes* 7(5).

Sodré D, Rocha T, Rendeiro F, Vallinoto M. 2018. A case of prolonged amplexus between males of *Rhinella mirandaribeiroi* and *R. marina* on Marajó Island, state of Pará, Brazil. *Herpetology Notes* 11, 127-128.

Toledo LF, Martins IA, Bruschi DP, Passos MA, Alexandr, C, Haddad CF. 2015. The anuran calling repertoire in the light of social context. *Acta ethologica* 18(2), 87-99.

Vásquez-Cruz V, Canseco-Márquez L, Pérez-Gámez EM. 2019. Two incidents of heterospecific amplexus involving Mexican treefrogs (*Smilisca baudini*), a Rio Grande leopard frog (*Lithobates berlandieri*), and a Morelet's leaf frog (*Agalychnis*

moreletii) (Amphibia: Anura: Hylidae, Ranidae, and Phyllomedusidae). *Reptiles & Amphibians Conservation and Natural History* 26 (2): 140-141.

Vega-Sánchez YM, Mendoza-Cuenca L, González-Rodríguez A. 2022. Morphological variation and reproductive isolation in the *Hetaerina americana* species complex. *Scientific reports* 12(1), 1-10.

Verrell PA. 1994. Is decreased frequency of mating among conspecifics a cost of sympatry in salamanders? *Evolution* 921-925.

Wells KD. 2007. The ecology and behavior of amphibians. Chicago: University of Chicago Press.

Wells KD. 1977. The social behaviour of anuran amphibians. *Animal Behaviour* 25, 666-693.

Winter AD, Rollenhagen T. 1990. The importance of male and female acoustic behaviour for reproductive isolation in *Ribautodelphax* planthoppers (Homoptera: Delphacidae). *Biological Journal of the Linnean Society* 40(2), 191-206.

Wyman MT, Charlton BD, Locatelli Y, Reby D. 2011. Variability of female responses to conspecific vs. heterospecific male mating calls in polygynous deer: an open door to hybridization? *PloS one* 6(8), e23296.

Zimić A, Ćurić A, Kolanek A, Bonk M, Herero D, Peyer N, Bühler C, Schmidt BR, Martinez N, Grossenbacher K, Jelić D. 2018. Making too many mistakes: cases of abnormal amplexi in european anurans. Poster presented at 2nd Balkan Herpetological Symposium within 13th Croatian Biological Congress with International Participation / 19th – 23rd September 2018 Poreč, Croatia

Caption to figures

Figure 1. Phylogenetic tree displaying the probability that misdirected amplexus occurred at each node and tip calculated using the HMM method. Genera of the species that have been observed in hetrospecific matings appear in blue font. Genera of the species that are unlikely to be involved in misdirected amplexus appear in red font. Genera for which this information was unknown appear in black font.

Figure 2. Accumulation through time of lineages in each of the mapped states (misdirected amplexus or not) using tip states estimated from the HMM method. This analysis dated the apparition of misdirected amplexus at the oldest node of the phylogenetic tree we used (220 Mya), while the loss of this character (i.e., the emergence of species that do not perform amplexus) was much more recent (60 Mya). The vertical dashed line represents the expected transition to no misdirected amplexus state.

Supplementary materials

Appendix A: List of species which have been observed in misdirected amplexus and corresponding literature sources. Only the species that mistakenly amplected other species were listed and used in our analyses.

Appendix B: References used for Appendix A.

Appendix C: Phylogenetic tree displaying the probability that misdirected amplexus occurred at each node and tip calculated using the HSP method. Genera of the species that have been observed in heterospecific matings appear in blue font. Genera of the species that are unlikely to be involved in misdirected amplexus appear in red font. Genera for which this information was unknown appear in black font.

Appendix D: Accumulation through time of lineages in each of the mapped states (misdirected amplexus or not) using tip states estimated from the HSP method. The vertical dashed line represents the expected transition to no misdirected amplexus state.

Appendix E: Number of studies reporting heterospecific misdirected amplexus in anuran amphibians relative to the year of publication. Statistical analysis (quasi-poisson GLM) indicate a recent exponential increase of misdirected amplexus observations (Estimate=0.0597, SE=0.0085, t-value=7.027, p-value<0.001).